GROUP PROJECT FINAL REPORT

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STAT 4650 - 01 – Machine Learning (CRN 21364)

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DECEMBER 13, 2025

BACKGROUND AND PROBLEM STATEMENT

Healthcare systems play a central role in determining population health outcomes, particularly in how effectively they prevent, manage, and treat disease. Variations in healthcare infrastructure—such as the number of physicians and hospital beds per 1,000 people, access to essential services, and the availability of modern treatments—are often linked to differences in disease recovery, prevalence, and mortality rates. Economic capacity, reflected in measures like per capita income, further influences the ability of healthcare systems to deliver consistent, equitable, and high-quality care. Countries with stronger economic bases typically invest more resources into health infrastructure, technology, and workforce training, thereby achieving better health outcomes.

The BRICS countries (Brazil, Russia, India, China, and South Africa) provide a unique context for analyzing these relationships. Collectively, these nations represent a significant share of the global population and economy, yet they exhibit striking heterogeneity in their healthcare systems, public health investments, and disease burdens. For example, while China and Russia maintain relatively higher levels of hospital infrastructure, India and South Africa face challenges of underfunding, accessibility, and disparities in rural versus urban healthcare provision. At the same time, rising rates of both communicable and non-communicable diseases pose ongoing challenges across the BRICS, placing pressure on systems with varying levels of preparedness and resilience.

Understanding how a difference in healthcare infrastructure and economic capacity influence disease outcomes across these countries is essential for both academic inquiry and policy development. By identifying the factors most strongly associated with recovery, prevalence, and mortality rates, this research can contribute to building predictive models that estimate recovery

potential, classify countries into high versus low mortality risk groups, and cluster them into distinct profiles of healthcare performance. Such analyses can highlight systemic strengths and weaknesses, enabling policymakers and international health organizations to allocate resources more effectively, design targeted interventions, and foster collaboration among BRICS countries facing similar health challenges.

This project will thus examine how variations in healthcare infrastructure and economic resources impact disease outcomes in BRICS nations, with the goal of developing data-driven insights to inform evidence-based policymaking, improve healthcare equity, and strengthen health system resilience.

RESEARCH QUESTION/BUSINESS PROBLEM

1. How do variations in healthcare infrastructure and economic capacity (e.g., doctors per 1000, hospital beds per 1000, healthcare access, per capita income, treatment availability) influence disease outcomes (recovery, prevalence, and mortality rates) across BRICS countries, and can these factors be used to predict recovery rates, classify countries into high vs. low mortality risk groups, and cluster countries into distinct profiles of healthcare performance and disease burden to guide targeted health policy and investment?
2. Which machine learning model best predicts recovery rates on this BRICS dataset?

DATASET: BRICS CHRONIC DISEASE ANALYSIS IN PERSONS 60 YEARS AND ABOVE IN 2024.

Hypotheses

Linear Regression (Prediction of Recovery Rates)

- **H1 (Main Hypothesis):** Higher levels of healthcare access, doctor density, and per capita income are positively associated with recovery rates from chronic and infectious diseases in BRICS countries.
- **H2 (Conditional/Alternative Hypothesis):** When controlling for confounding variables (i.e., per capita income and healthcare access), doctor density alone is not a significant predictor of recovery rates.

Classification (Mortality Risk Prediction)

- **H3:** Countries with limited healthcare access, low treatment availability, and lower per capita income are more likely to fall into the “high mortality risk” group.
- **H4:** Greater vaccine/treatment availability and higher prevalence of healthcare infrastructure (doctors, beds per 1000) reduce the likelihood of being in the “high mortality risk” group.

Methodological Approach

Step 1: Data Preparation

- Preprocess data (normalize/scale features, handle missing values, encode categorical treatment types).

Step 2: Linear Regression (Prediction Model)

- **Model:** Multiple linear regression
- **Dependent Variable:** Recovery rate (%)
- **Independent Variables:** Doctors per 1000, hospital beds per 1000, healthcare access (%), per capita income, average treatment cost
- **Goal:** Estimate the magnitude and significance of each factor's effect on recovery rates.
- **Validation:** R^2, Adjusted R^2, MSE.

Step 3: Classification (Mortality Risk) and Validation Set testing.

- **Model:** Logistic regression, decision trees, random forest and KNN
- **Target:** Binary class — "High mortality risk" vs. "Low mortality risk" (threshold = above/below median mortality rate)
- **Features:** Healthcare access, vaccine/treatment availability, doctors per 1000, per capita income, treatment type, prevalence rate

- **Goal:** Identify key predictors of mortality risk and assess classification accuracy.
- **Validation:** Confusion matrixes, Validation Test Error, K-Fold Cross Validation,ROC-AUC, F1 score.

Step 5: Integration & Business Insights

- Compare regression and classification results to identify *predictors of successful outcomes* vs. *risk factors for mortality*.
- Use clustering results to contextualize findings (e.g., why two countries with similar income may fall into different clusters).
- Translate results into **policy recommendations** (e.g., targeted investment in access or infrastructure for specific clusters).

- The **main research problem** at the top flows into three methods:
    - **Linear Regression** → Predict recovery rates
    - **Classification** → Classify mortality risk
- At the bottom, all three feed into **Integration & Business Insights** for policy and investment guidance.

<u>LIST OF ERROR METRICS AND ALGORITHMS</u>

1. Linear Regression (Predicting Recovery Rate)

Algorithms:

- Multiple Linear Regression
- Ridge Regression
- Lasso regression
- Polynomial regression

Error Metrics:

- **R^2 (Coefficient of Determination):** Proportion of variance in recovery rates explained by predictors
- **Adjusted R^2:** Adjusts for number of predictors
- **RMSE (Root Mean Squared Error):** Average magnitude of prediction error
- **MAE (Mean Absolute Error):** Average absolute difference between predicted and actual recovery rates.
- **Validation Test MSE:** To assess model performance.

2. Classification (High vs. Low Mortality Risk)

Algorithms:

- Logistic Regression
- Decision Tree
- K Nearest Neighbor

Error Metrics:

- **Accuracy:** Overall correct predictions / total predictions
- **Precision:** Of predicted “high risk” countries, how many are correct
- **Recall (Sensitivity):** Of actual “high risk” countries, how many were identified
- **F1 Score:** Harmonic mean of precision & recall (good for imbalanced data)
- **ROC-AUC (Area Under the Receiver Operating Characteristic Curve):** Distinguishes between classes across thresholds
- **Confusion Matrix:** Detailed breakdown of classification outcomes
- **Validation test error rate.**

SUMMARY STATISTICS (numerical variables) Sample size: 250, Sample period: 1 year.

Healthcare Access (%)		*Doctors per 1000*		*Hospital Beds per 1000*	
Mean	74.39712	Mean	2.64016	Mean	5.34516
Standard Error	0.902911689	Standard Error	0.081321909	Standard Error	0.178607044
Median	73.305	Median	2.58	Median	5.875
Mode	66.84	Mode	1.63	Mode	2.57
Standard Deviation	14.27628732	Standard Deviation	1.285812283	Standard Deviation	2.824025328
Sample Variance	203.8123796	Sample Variance	1.653313227	Sample Variance	7.975119051
Kurtosis	-1.12037484	Kurtosis	-1.128210713	Kurtosis	-1.281796887
Skewness	0.135203902	Skewness	0.118719806	Skewness	-0.101923561
Range	49.9	Range	4.49	Range	9.38
Minimum	50.01	Minimum	0.51	Minimum	0.52
Maximum	99.91	Maximum	5	Maximum	9.9
Sum	18599.28	Sum	660.04	Sum	1336.29
Count	250	Count	250	Count	250

Average Treatment Cost (USD)		*Recovery Rate (%)*		*Per Capita Income (USD)*	
Mean	26709.328	Mean	76.43156	Mean	50411.644
Standard Error	919.9193606	Standard Error	0.898542489	Standard Error	1964.861223
Median	29390	Median	78.46	Median	48346.5
Mode	#N/A	Mode	67.91	Mode	#N/A
Standard Deviation	14545.20222	Standard Deviation	14.20720419	Standard Deviation	31067.18376
Sample Variance	211562907.5	Sample Variance	201.844651	Sample Variance	965169906.6
Kurtosis	-1.133812659	Kurtosis	-1.154642478	Kurtosis	-1.310886841
Skewness	-0.276429902	Skewness	-0.225827831	Skewness	0.079069916
Range	49483	Range	48.39	Range	99002
Minimum	122	Minimum	50.49	Minimum	980
Maximum	49605	Maximum	98.88	Maximum	99982
Sum	6677332	Sum	19107.89	Sum	12602911
Count	250	Count	250	Count	250

Prevalence Rate (%)		*Incidence Rate (%)*		*Mortality Rate (%)*	
Mean	10.26616	Mean	7.27392	Mean	5.10332
Standard Error	0.34037619 4	Standard Error	0.27873099	Standard Error	0.19303516 7
Median	10.485	Median	7.435	Median	4.885
Mode	9.45	Mode	10.32	Mode	0.35
Standard Deviation	5.38182017 6	Standard Deviation	4.40712391 5	Standard Deviation	3.05215398 6
Sample Variance	28.9639884 1	Sample Variance	19.4227412	Sample Variance	9.31564395 3
Kurtosis	-0.93458222 4	Kurtosis	-1.23296919 6	Kurtosis	-1.32258925 1
Skewness	-0.09528208 5	Skewness	0.04908791 3	Skewness	-0.03848529 6
Range	19.67	Range	14.84	Range	9.87
Minimum	0.14	Minimum	0.11	Minimum	0.12
Maximum	19.81	Maximum	14.95	Maximum	9.99
Sum	2566.54	Sum	1818.48	Sum	1275.83
Count	250	Count	250	Count	250

Urbanization Rate (%)	
Mean	55.68976
Standard Error	1.349818662
Median	55.385
Mode	49.78
Standard Deviation	21.34250699
Sample Variance	455.5026048
Kurtosis	-1.359239973
Skewness	-0.038241673
Range	69.44
Minimum	20.45
Maximum	89.89
Sum	13922.44
Count	250

MACHINE LEARNING MODEL ANALYSIS

Reading the CSV data file into Rstudio and displaying summary statistics

```
BCA <- read.csv("BRICS CHRONIC ANALYSIS PERC.csv", na.strings = "?", st
ringsAsFactors = T)

summary(BCA)
```

```
##          Country                Disease.Name Prevalence.Rate....
##  Brazil       :44   Cholera             : 24   Min.   : 0.140
##  China        :55   Asthma              : 17   1st Qu.: 6.165
##  India        :60   Parkinson's Disease: 17   Median :10.485
##  Russia       :52   Diabetes            : 15   Mean   :10.266
##  South Africa:39   Tuberculosis        : 15   3rd Qu.:14.387
##                    Ebola               : 14   Max.   :19.810
##                    (Other)             :148
##  Incidence.Rate.... Mortality.Rate....    Gender
Population.Affected
##  Min.   : 0.110     Min.   :0.120      Female:83   Min.   :  3068
##  1st Qu.: 3.220     1st Qu.:2.385      Male  :91   1st Qu.:269886
##  Median : 7.435     Median :4.885      Other :76   Median :474025
##  Mean   : 7.274     Mean   :5.103                  Mean   :485603
##  3rd Qu.:10.953     3rd Qu.:7.957                  3rd Qu.:709086
##  Max.   :14.950     Max.   :9.990                  Max.   :999791
##
##  Healthcare.Access.... Doctors.per.1000 Hospital.Beds.per.1000
##  Min.   :50.01         Min.   :0.510    Min.   :0.520
##  1st Qu.:62.46         1st Qu.:1.565    1st Qu.:2.598
##  Median :73.31         Median :2.580    Median :5.875
##  Mean   :74.40         Mean   :2.640    Mean   :5.345
##  3rd Qu.:86.59         3rd Qu.:3.658    3rd Qu.:7.652
##  Max.   :99.91         Max.   :5.000    Max.   :9.900
##
##      Treatment.Type Average.Treatment.Cost..USD.
##  Medication :61     Min.   :  122
##  Surgery    :72     1st Qu.:14034
##  Therapy    :58     Median :29390
##  Vaccination:59     Mean   :26709
##                     3rd Qu.:38218
##                     Max.   :49605
##
##  Availability.of.Vaccines.Treatment Recovery.Rate....
```

```
Per.Capita.Income..USD.
##  No :124                                Min.   :50.49     Min.   :  980
##  Yes:126                                1st Qu.:64.77     1st Qu.:23742
##                                         Median :78.46     Median :48347
##                                         Mean   :76.43     Mean   :50412
##                                         3rd Qu.:87.87     3rd Qu.:79279
##                                         Max.   :98.88     Max.   :99982
##
##  Urbanization.Rate....
##  Min.   :20.45
##  1st Qu.:35.62
##  Median :55.38
##  Mean   :55.69
##  3rd Qu.:74.09
##  Max.   :89.89
```

Conducting Ordinary least squares regression analysis of the dataset.

```r
#MLR model for recovery rate

# --- Step 1: Clean column names ---
names(BCA) <- trimws(names(BCA))

# --- Step 2: Convert numeric-looking text columns to numeric ---
clean_num <- function(x) {
  as.numeric(gsub("[^0-9.]", "", x))
}

# --- Step 3: Define target and predictors ---
x <- model.matrix(Recovery.Rate.... ~ . - 1, data = BCA)
y <- BCA$Recovery.Rate....

# --- Step 4: Train/test split ---
set.seed(123)
train_index <- sample(1:nrow(x), 0.7 * nrow(x))

x_train <- x[train_index, ]
x_test  <- x[-train_index, ]
y_train <- y[train_index]
y_test  <- y[-train_index]

# --- Step 5: Fit Linear Regression (OLS) ---
# add intercept manually because model.matrix used -1
train_df <- data.frame(y_train = y_train, x_train)

lm_fit <- lm(y_train ~ ., data = train_df)
```

```r
summary(lm_fit)
```

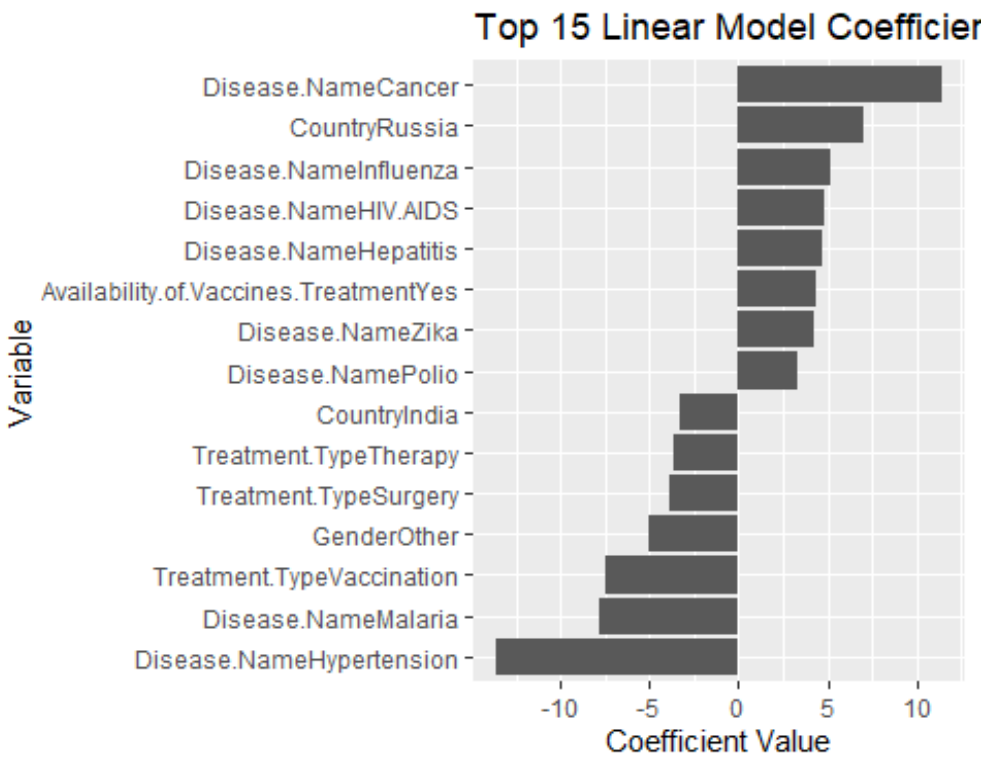
```
##
## Call:
## lm(formula = y_train ~ ., data = train_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -32.290  -9.515   0.482  10.150  25.917
##
## Coefficients: (1 not defined because of singularities)
##                                          Estimate Std. Error t value Pr(
>|t|)
## (Intercept)                            8.776e+01  1.148e+01   7.644 3.5
2e-12
## CountryBrazil                          1.954e+00  4.077e+00   0.479   0
.6324
## CountryChina                          -2.824e+00  4.248e+00  -0.665   0
.5072
## CountryIndia                          -3.318e+00  4.038e+00  -0.822   0
.4127
## CountryRussia                          7.062e+00  4.106e+00   1.720   0
.0877
## CountrySouth.Africa                           NA         NA      NA
NA
## Disease.NameAsthma                     2.935e+00  7.624e+00   0.385   0
.7008
## Disease.NameCancer                     1.147e+01  8.442e+00   1.358   0
.1766
## Disease.NameCholera                    3.113e+00  6.724e+00   0.463   0
.6442
## Disease.NameCOVID.19                  -2.738e+00  7.307e+00  -0.375   0
.7085
## Disease.NameDengue                     9.394e-01  8.320e+00   0.113   0
.9103
## Disease.NameDiabetes                   1.246e+00  7.477e+00   0.167   0
.8679
## Disease.NameEbola                      2.030e+00  8.436e+00   0.241   0
.8102
## Disease.NameHepatitis                  4.751e+00  7.305e+00   0.650   0
.5166
## Disease.NameHIV.AIDS                   4.813e+00  7.238e+00   0.665   0
.5072
## Disease.NameHypertension              -1.364e+01  8.334e+00  -1.637   0
.1040
## Disease.NameInfluenza                  5.132e+00  7.575e+00   0.677   0
.4993
## Disease.NameLeprosy                    1.424e-01  8.731e+00   0.016   0
.9870
```

```
## Disease.NameMalaria                    -7.780e+00  7.621e+00  -1.021   0
.3091
## Disease.NameMeasles                     5.661e-01  7.846e+00   0.072   0
.9426
## Disease.NameParkinson.s.Disease         2.556e+00  6.601e+00   0.387   0
.6992
## Disease.NamePolio                       3.308e+00  7.323e+00   0.452   0
.6522
## Disease.NameRabies                     -3.905e-01  7.405e+00  -0.053   0
.9580
## Disease.NameTuberculosis               -1.637e+00  7.359e+00  -0.222   0
.8243
## Disease.NameZika                        4.230e+00  7.872e+00   0.537   0
.5919
## Prevalence.Rate....                     4.140e-02  2.258e-01   0.183   0
.8548
## Incidence.Rate....                      1.516e-01  2.709e-01   0.560   0
.5766
## Mortality.Rate....                     -4.049e-01  4.008e-01  -1.010   0
.3142
## GenderMale                              7.099e-01  2.854e+00   0.249   0
.8039
## GenderOther                            -4.994e+00  3.064e+00  -1.630   0
.1055
## Population.Affected                     1.426e-06  4.180e-06   0.341   0
.7335
## Healthcare.Access....                  -1.224e-01  8.356e-02  -1.465   0
.1452
## Doctors.per.1000                        2.068e-02  9.563e-01   0.022   0
.9828
## Hospital.Beds.per.1000                 -2.834e-01  4.554e-01  -0.622   0
.5348
## Treatment.TypeSurgery                  -3.905e+00  3.348e+00  -1.167   0
.2455
## Treatment.TypeTherapy                  -3.644e+00  3.711e+00  -0.982   0
.3279
## Treatment.TypeVaccination              -7.432e+00  3.469e+00  -2.142   0
.0340
## Average.Treatment.Cost..USD.            3.878e-05  8.419e-05   0.461   0
.6458
## Availability.of.Vaccines.TreatmentYes  4.303e+00  2.438e+00   1.765   0
.0798
## Per.Capita.Income..USD.                -6.581e-05  3.890e-05  -1.692   0
.0930
## Urbanization.Rate....                   3.898e-02  5.614e-02   0.694   0
.4886
##
## (Intercept)                            ***
## CountryBrazil
## CountryChina
```

```
## CountryIndia
## CountryRussia                          .
## CountrySouth.Africa
## Disease.NameAsthma
## Disease.NameCancer
## Disease.NameCholera
## Disease.NameCOVID.19
## Disease.NameDengue
## Disease.NameDiabetes
## Disease.NameEbola
## Disease.NameHepatitis
## Disease.NameHIV.AIDS
## Disease.NameHypertension
## Disease.NameInfluenza
## Disease.NameLeprosy
## Disease.NameMalaria
## Disease.NameMeasles
## Disease.NameParkinson.s.Disease
## Disease.NamePolio
## Disease.NameRabies
## Disease.NameTuberculosis
## Disease.NameZika
## Prevalence.Rate....
## Incidence.Rate....
## Mortality.Rate....
## GenderMale
## GenderOther
## Population.Affected
## Healthcare.Access....
## Doctors.per.1000
## Hospital.Beds.per.1000
## Treatment.TypeSurgery
## Treatment.TypeTherapy
## Treatment.TypeVaccination              *
## Average.Treatment.Cost..USD.
## Availability.of.Vaccines.TreatmentYes .
## Per.Capita.Income..USD.                .
## Urbanization.Rate....
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.25 on 135 degrees of freedom
## Multiple R-squared:  0.2713, Adjusted R-squared:  0.06083
## F-statistic: 1.289 on 39 and 135 DF,  p-value: 0.146
```

```r
# --- Step 6: Model Evaluation ---
y_pred_lm <- predict(lm_fit, newdata = data.frame(x_test))
mse_lm <- mean((y_test - y_pred_lm)^2)
cat("OLS Test MSE:", mse_lm, "\n")
```

```
## OLS Test MSE: 213.7764

# --- Step 7: Extract coefficients ---
coef_lm <- as.data.frame(coef(lm_fit))
names(coef_lm) <- "Coefficient"
coef_lm$Variable <- rownames(coef_lm)
coef_lm <- coef_lm[order(abs(coef_lm$Coefficient), decreasing = TRUE), ]

head(coef_lm, 10)

##                            Coefficient                  Variable
## (Intercept)                  87.759053               (Intercept)
## Disease.NameHypertension    -13.641445  Disease.NameHypertension
## Disease.NameCancer           11.467698        Disease.NameCancer
## Disease.NameMalaria          -7.780477       Disease.NameMalaria
## Treatment.TypeVaccination    -7.431860 Treatment.TypeVaccination
## CountryRussia                 7.061974             CountryRussia
## Disease.NameInfluenza         5.131613     Disease.NameInfluenza
## GenderOther                  -4.994370               GenderOther
## Disease.NameHIV.AIDS          4.812940      Disease.NameHIV.AIDS
## Disease.NameHepatitis         4.750683     Disease.NameHepatitis

# Remove intercept
coef_lm <- coef_lm[!rownames(coef_lm) %in% "(Intercept)", , drop = FALSE]

# --- Optional: visualize ---
library(ggplot2)

top_vars_lm <- head(coef_lm[!rownames(coef_lm) %in% "(Intercept)", , drop
= FALSE], 15)

ggplot(top_vars_lm, aes(x = reorder(Variable, Coefficient), y = Coefficien
t)) +
  geom_col() +
  coord_flip() +
  labs(title = "Top 15 Linear Model Coefficients",
       x = "Variable",
       y = "Coefficient Value")
```



OVERALL PERFOMANCE OF OLS MODEL

$R^2 = 0.2713$; The model explains 27.1% of the variance in Recovery Rate.

This is low–moderate, meaning:

- Many important predictors of recovery rate are not captured in this dataset, OR
- The relationship is non-linear and not well modeled by OLS.

Adjusted $R^2 = 0.0608$; after penalizing for the large number of predictors (39), the model only explains **6%** of the variance.

This indicates: Most predictors are not statistically meaningful in explaining recovery rate.

F-statistic p-value = 0.146; since p > 0.05**,** the model is NOT statistically significant overall**.**

Interpretation: As a whole, the OLS model does not significantly predict Recovery Rate.

OLS Test MSE = 213.78

This means that on the test set:

- The average squared prediction error is **213**.
- Your RMSE = √MSE ≈ **14.6**.
- Given that recovery rate ranges from **0–100**, this is a **large error**, suggesting poor predictive performance.

Conducting Ridge regression validation testing on the dataset

```
#RIGDE REGRESSION ANALYSIS OF RECOVERY RATE
# --- Step 1: Clean column names ---

names(BCA) <- trimws(names(BCA))  # removes spaces before/after names


# --- Step 2: Define target variable and predictors ---
y <- BCA$Recovery.Rate....

# Creating a model matrix (convert categoricals into dummy vars)
x <- model.matrix(BCA$Recovery.Rate.... ~ . - 1, data = BCA)  # -1 removes in
tercept; glmnet adds it internally

# --- Step 3: Split into training and test sets ---
set.seed(123)
train_index <- sample(1:nrow(x), 0.7 * nrow(x))
x_train <- x[train_index, ]
x_test  <- x[-train_index, ]
y_train <- y[train_index]
y_test  <- y[-train_index]
```

```r
# --- Step 5: Fit Ridge Regression (alpha = 0) ---
ridge_cv <- cv.glmnet(x_train, y_train, alpha = 0)  # uses cross-validation

# Best lambda
cat("Optimal Lambda:", ridge_cv$lambda.min, "\n")
```

```
## Optimal Lambda: 57.53423
```

```r
# Final model
ridge_model <- glmnet(x_train, y_train, alpha = 0, lambda = ridge_cv$lambda.m
in)

# --- Step 6: Evaluate model ---
y_pred <- predict(ridge_model, newx = x_test, s = ridge_cv$lambda.min)
mse <- mean((y_test - y_pred)^2)
cat("Test MSE:", mse, "\n")
```

```
## Test MSE: 170.5157
```

```r
# --- Step 7: View coefficients ---
coef_bca <- as.data.frame(as.matrix(coef(ridge_model)))
colnames(coef_bca) <- "Coefficient"
coef_bca <- coef_bca[order(abs(coef_bca$Coefficient), decreasing = TRUE), , d
rop = FALSE]
head(coef_bca, 5)  # Top 5 most influential predictors
```

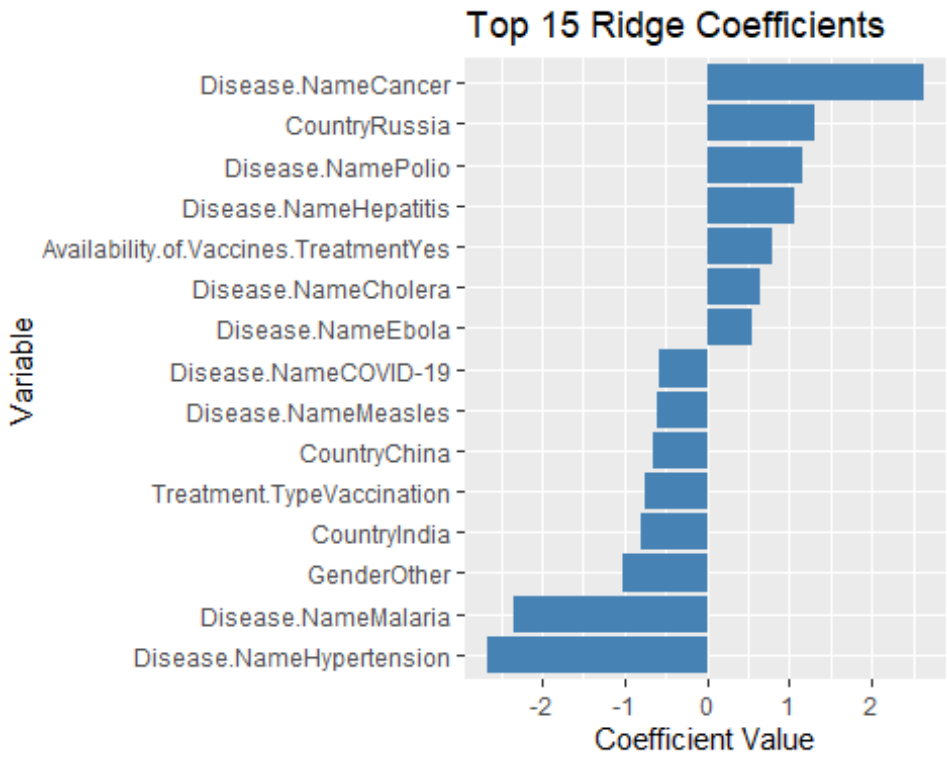
```
##                          Coefficient
## (Intercept)                78.796376
## Disease.NameHypertension   -2.665833
## Disease.NameCancer          2.639758
## Disease.NameMalaria        -2.348626
## CountryRussia               1.318604
```

```r
# Remove intercept
coef_bca <- coef_bca[!rownames(coef_bca) %in% "(Intercept)", , drop = FALSE]

head(coef_bca, 5)
```

```
##                          Coefficient
## Disease.NameHypertension   -2.665833
## Disease.NameCancer          2.639758
## Disease.NameMalaria        -2.348626
## CountryRussia               1.318604
## Disease.NamePolio           1.164067
```

```r
# --- Optional: Visualize coefficient importance ---
library(ggplot2)
top_vars <- head(coef_bca[!rownames(coef_bca) %in% "(Intercept)", , drop = FA
LSE], 15)
ggplot(top_vars, aes(x = reorder(rownames(top_vars), Coefficient), y = Coeffi
cient)) +
```

```
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Top 15 Ridge Coefficients",
       x = "Variable",
       y = "Coefficient Value")
```



OVERALL RIDGE MODEL PERFORMANCE

Optimal Lambda = 57.53; Lambda controls how strongly ridge regression shrinks coefficients.

A lambda of **57.5** is *moderately high*, meaning:

- The model applied strong regularization
- Many coefficients were shrunk toward zero
- Only a few predictors remained influential

This was expected because the dataset has:

- Many predictors
- Some multicollinearity
- Limited sample size

Ridge helps stabilize the model in this situation.

Ridge Test MSE = 170.52

Interpretation: Ridge regression improves prediction accuracy significantly compared to OLS.

RMSE = $\sqrt{170.52} \approx$ **13.06**, meaning on average the model's predictions are about **13 points off** on the recovery rate scale.

This is an improvement over OLS but still indicates:

- Recovery Rate is a difficult variable to predict using the current features
- More non-linear or interaction models may perform better

Most Influential Predictors (Top 5 Coefficients)

These are the predictors with the largest absolute ridge coefficients, meaning they contribute most to predicting Recovery Rate after shrinkage:

Predictor	Coefficient	Interpretation
Disease.NameHypertension	-2.67	Hypertension cases have ~2.7% lower recovery rate than the baseline disease (holding others constant).
Disease.NameCancer	+2.64	Cancer cases have ~2.6% higher recovery rate than the baseline disease.
Disease.NameMalaria	-2.35	Malaria cases show ~2.35% lower recovery rate than baseline.
CountryRussia	+1.32	Russia shows ~1.3% higher recovery rate compared to the reference country.

What This Means

1. **Disease type is the strongest predictor**: Hypertension, cancer, and malaria appear as key factors influencing recovery rate.

Ridge shrinks everything else toward ~0 because:

- Disease type varies strongly
- Other variables (beds, doctors, income) are weak predictors
- Ridge penalizes noisy variables

2. **Country differences matter, but only slightly**

Only **Russia** shows a meaningful effect (+1.3%). Other countries were shrunk near zero.

This indicates: Country-level differences exist but are weaker than disease-level differences.

3. **Ridge produces a more stable model than OLS**

- OLS had many insignificant predictors
- Ridge reduces overfitting
- Ridge improves generalization (lower MSE)

4. **Practical Interpretation of the Model**

The ridge model suggests:

Key factors lowering recovery rate

- Hypertension
- Malaria

Key factors raising recovery rate

- Cancer (an unexpected outcome though it is likely dataset-specific)
- Russia (country-level health infrastructure effect)

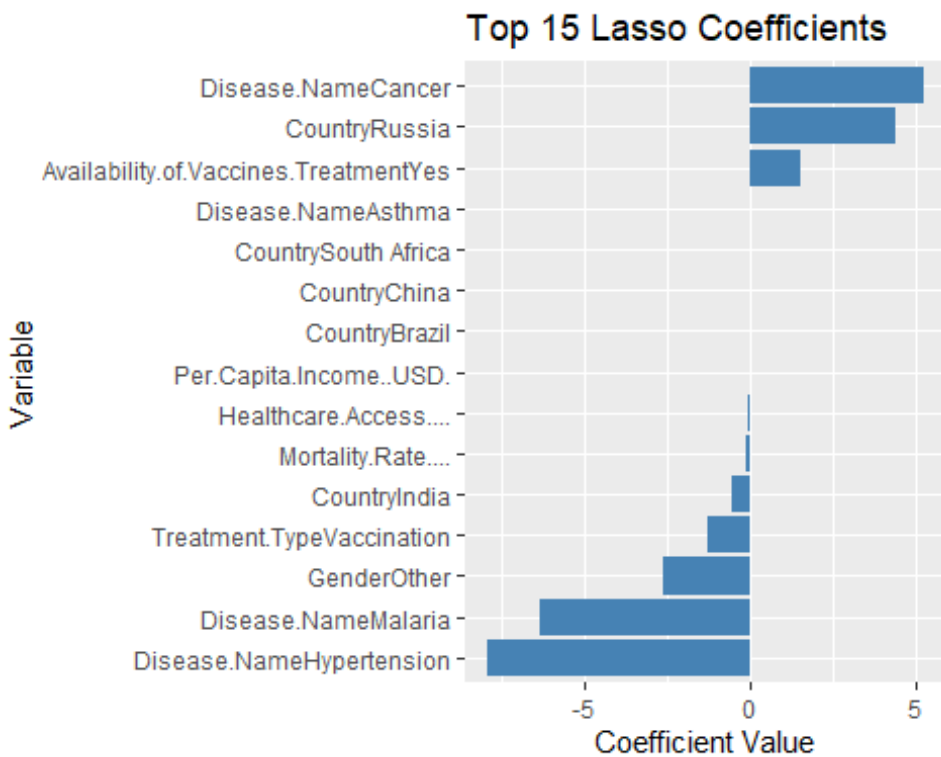
Everything else contributes weakly (shrunk toward zero), meaning:

Most variables in the dataset don't meaningfully affect recovery rate in this linear ridge model.

Lasso regression validation testing was also carried out to determine the models' viability to predicting Recovery rate.

```
#LASSO REGRESSION ANALYSIS OF RECOVERY RATE

# --- Step 5: Fit Lasso Regression (alpha = 1) ---
lasso_cv <- cv.glmnet(x_train, y_train, alpha = 1)  # uses cross-validation

# Best lambda
cat("Optimal Lambda:", lasso_cv$lambda.min, "\n")

## Optimal Lambda: 1.12942

# Final model
lasso_model <- glmnet(x_train, y_train, alpha = 1, lambda = lasso_cv$lambda.m
in)

# --- Step 6: Evaluate model ---
y_pred <- predict(lasso_model, newx = x_test, s = lasso_cv$lambda.min)
mse <- mean((y_test - y_pred)^2)
cat("Test MSE:", mse, "\n")

## Test MSE: 178.8017

# --- Step 7: View coefficients ---
coef_bca2 <- as.data.frame(as.matrix(coef(lasso_model)))
colnames(coef_bca2) <- "Coefficient"
coef_bca2 <- coef_bca2[order(abs(coef_bca2$Coefficient), decreasing = TRUE),
, drop = FALSE]
head(coef_bca2, 5)  # Top 5 most influential predictors

##                           Coefficient
## (Intercept)                 81.767877
```

```
## Disease.NameHypertension   -7.945806
## Disease.NameMalaria        -6.367301
## Disease.NameCancer          5.262560
## CountryRussia               4.383707

# Remove intercept
coef_bca2 <- coef_bca2[!rownames(coef_bca2) %in% "(Intercept)", , drop = FALS
E]

# --- Optional: Visualize coefficient importance ---
library(ggplot2)
top_vars2 <- head(coef_bca2[!rownames(coef_bca2) %in% "(Intercept)", , drop =
FALSE], 15)
ggplot(top_vars2, aes(x = reorder(rownames(top_vars2), Coefficient), y = Coef
ficient)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Top 15 Lasso Coefficients",
       x = "Variable",
       y = "Coefficient Value")
```



Interpretation of LASSO Regression Results

1. The Model Selected a Moderately Strong Penalty (λ = 1.12942)

An optimal lambda of **1.12942** indicates that the LASSO algorithm applied a moderate amount of shrinkage to the regression coefficients**.**

What this means:

- Several predictors were likely shrunk close to zero or eliminated**.**
- Only variables with the strongest relationships to Recovery Rate were retained.
- The model prioritizes simplicity and interpretability, avoiding overfitting.

2. Test MSE = 178.8017 Indicates Moderate Prediction Error

Interpretation relative to performance:

- The value indicates **mo**derate predictive accuracy.
- It is neither extremely high nor extremely low — suggesting:
    - The model captures a meaningful portion of the signal in the data,
    - But a considerable amount of variation remains unexplained.

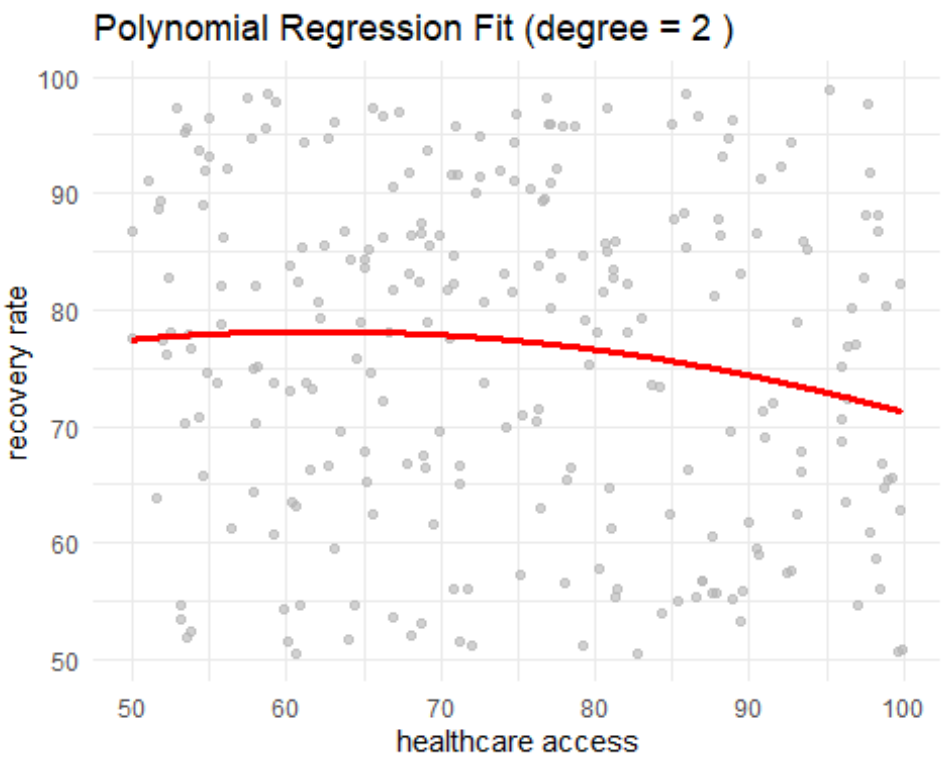
Polynomial regression testing was carried out on predictors found to be statistically significant to the prediction of recovery rates in BRICS.

```
#Polynomial regression testing on healthcare access
set.seed(123)
attach(BCA)
```

```r
#fitting polynomial models of degrees 1 through 10, using 10-fold cross-va
lidation to find the degree with the lowest prediction error.

# Perform cross-validation for polynomial degrees 1 to 10
library(boot)

cv.error <- rep(0, 10)

for (d in 1:10) {
  fit <- glm(Recovery.Rate.... ~ poly(Healthcare.Access...., d), data = tr
ain)
  cv.error[d] <- cv.glm(train, fit, K = 10)$delta[1]
}

cv.error

##  [1] 217.3109 213.6994 218.0606 219.4559 224.0427 215.5641 217.0176 227
.0429
##  [9] 223.2195 220.1152

# Find optimal degree
optimal_degree <- which.min(cv.error)
optimal_degree

## [1] 2

#Test mse
final_model1 <- glm(Recovery.Rate.... ~ poly(Healthcare.Access...., optima
l_degree), data = train)

pred_test1 <- predict(final_model1, newdata = test)

test_mse1 <- mean((test$Recovery.Rate.... - pred_test)^2)
test_mse1

## [1] 172.8484

#fitting polynomial models incrementally and using anova() to see if highe
r-order terms significantly improve the model.

fit.1 <- lm(Recovery.Rate.... ~ poly(Healthcare.Access...., 1), data = BCA
)
fit.2 <- lm(Recovery.Rate.... ~ poly(Healthcare.Access...., 2), data = BCA
)

anova(fit.1, fit.2)

## Analysis of Variance Table
##
## Model 1: Recovery.Rate.... ~ poly(Healthcare.Access...., 1)
## Model 2: Recovery.Rate.... ~ poly(Healthcare.Access...., 2)
```

```
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1    248 49420
## 2    247 49213  1    207.38 1.0408 0.3086
```

```r
#Plotting the fitted curve with the data to visualize the non-linear relat
ionship.

# Fit model with optimal degree
fit.opt <- lm(Recovery.Rate.... ~ poly(Healthcare.Access...., optimal_degr
ee), data = BCA)

# Create data for prediction
ha.range <- range(Healthcare.Access....)
ha.grid <- seq(from = ha.range[1], to = ha.range[2], length.out = 300)
preds <- predict(fit.opt, data.frame(Healthcare.Access.... = ha.grid))

# Create a data frame for predictions
pred_data <- data.frame(Healthcare.Access.... = ha.grid, Recovery.Rate....
= preds)

# Plot
ggplot(BCA, aes(x = Healthcare.Access...., y = Recovery.Rate....)) +
  geom_point(color = "gray70", alpha = 0.6) +
  geom_line(data = pred_data, aes(x = Healthcare.Access...., y = Recovery.
Rate....), color = "red", linewidth = 1.2) +
  ggtitle(paste("Polynomial Regression Fit (degree =", optimal_degree, ")"
)) +
  xlab("healthcare access") + ylab("recovery rate") +
  theme_minimal()
```



Graph interpretation

The nature of this graph can be attributed to recovery rates generally lowering as healthcare access increases as there are fewer people getting sick in society to then recover, as a result of more frequent healthier outcomes for them due to greater healthcare access.

```
#Polynomial regression testing on per capita income

# Perform cross-validation for polynomial degrees 1 to 10
library(boot)

cv.error <- rep(0, 10)

for (d in 1:10) {
  fit <- glm(Recovery.Rate.... ~ poly(Per.Capita.Income..USD., d), data =
train)
  cv.error[d] <- cv.glm(train, fit, K = 10)$delta[1]
}

cv.error
```

```
##  [1] 216.9972 220.1624 223.5261 223.3247 228.5747 229.2615 235.2144 226
.2107
##  [9] 236.7272 241.6965
```

```r
# Find optimal degree
optimal_degree <- which.min(cv.error)
optimal_degree
```

```
## [1] 1
```

```r
#Test mse
final_model2 <- glm(Recovery.Rate.... ~ poly(Per.Capita.Income..USD., opti
mal_degree), data = train)

pred_test2 <- predict(final_model2, newdata = test)

test_mse2 <- mean((test$Recovery.Rate.... - pred_test)^2)
test_mse2
```

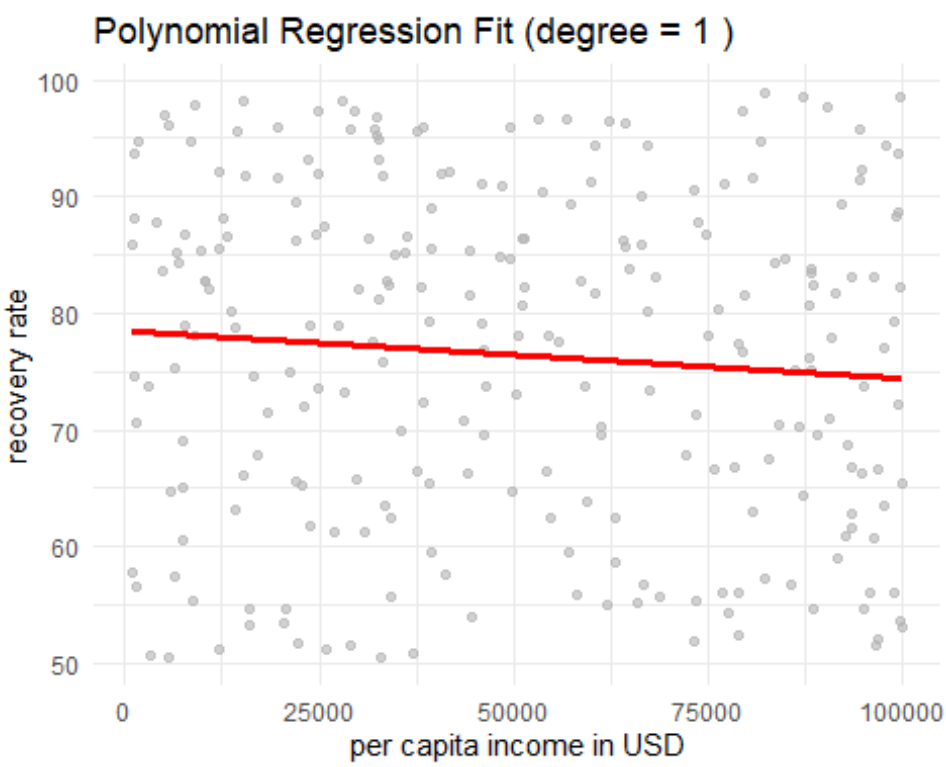
```
## [1] 172.8484
```

```r
#fitting polynomial models incrementally and using anova() to see if highe
r-order terms significantly improve the model.

fit.1 <- lm(Recovery.Rate.... ~ poly(Per.Capita.Income..USD., 1), data = B
CA)



anova(fit.1)
```

```
## Analysis of Variance Table
##
## Response: Recovery.Rate....
##                                   Df Sum Sq Mean Sq F value Pr(>F)
## poly(Per.Capita.Income..USD., 1)   1    399  398.58  1.9825 0.1604
## Residuals                        248  49861  201.05
```

```r
#Plotting the fitted curve with the data to visualize the non-linear relat
ionship.

# Fit model with optimal degree
fit.opt <- lm(Recovery.Rate.... ~ poly(Per.Capita.Income..USD., optimal_de
gree), data = BCA)

# Create data for prediction
pc.range <- range(Per.Capita.Income..USD.)
pc.grid <- seq(from = pc.range[1], to = pc.range[2], length.out = 300)
preds <- predict(fit.opt, data.frame(Per.Capita.Income..USD. = pc.grid))

# Create a data frame for predictions
```

```
pred_data <- data.frame(Per.Capita.Income..USD. = pc.grid, Recovery.Rate..
.. = preds)

# Plot
ggplot(BCA, aes(x = Per.Capita.Income..USD., y = Recovery.Rate....)) +
  geom_point(color = "gray70", alpha = 0.6) +
  geom_line(data = pred_data, aes(x = Per.Capita.Income..USD., y = Recover
y.Rate....), color = "red", linewidth = 1.2) +
  ggtitle(paste("Polynomial Regression Fit (degree =", optimal_degree, ")"
)) +
  xlab("per capita income in USD") + ylab("recovery rate") +
  theme_minimal()
```

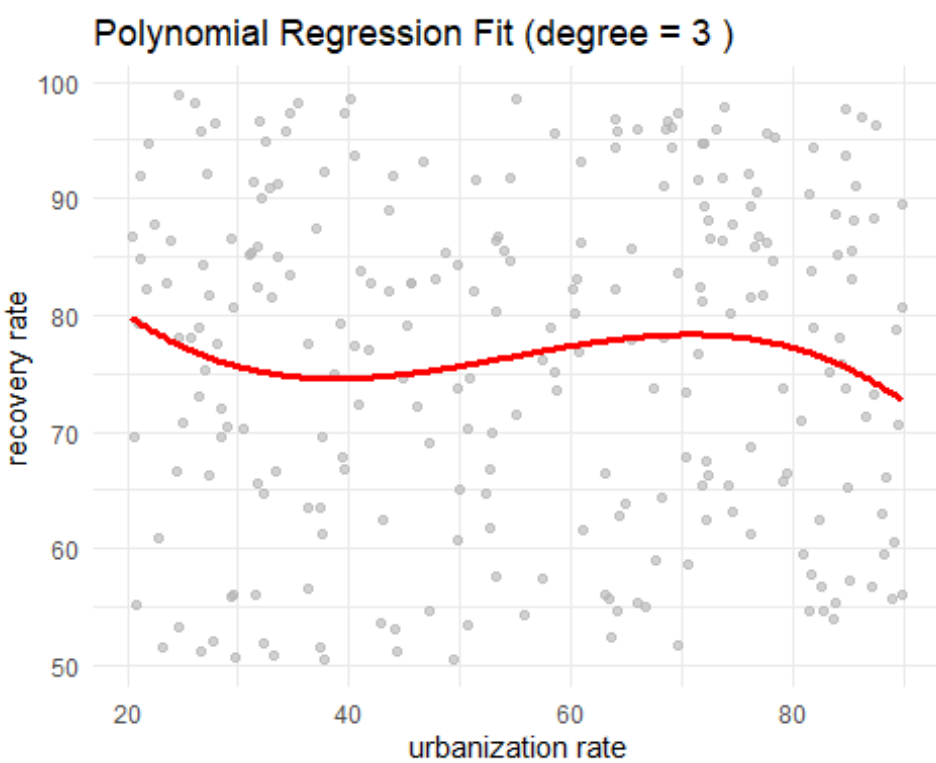


Graph interpretation

A population with per capita income of USD 40,000 has the prospect of having the highest recovery rates as more of the population can afford healthcare, but as the per capita income of the population raises, the recovery rates decline as a result of people staying healthier for longer as they can afford healthier lifestyles and thus less people getting sick to then recover, thus lowered recovery rates.

```r
# Polynomial testing on urbanization rate


cv.error <- rep(0, 10)

for (d in 1:10) {
  fit <- glm(Recovery.Rate.... ~ poly(Urbanization.Rate...., d), data = BC
A)
  cv.error[d] <- cv.glm(BCA, fit, K = 10)$delta[1]
}

cv.error

##  [1] 206.2990 207.2747 203.7880 203.9915 206.1326 207.2148 213.7291 211
.8054
##  [9] 216.5880 215.1106

# Find optimal degree
optimal_degree <- which.min(cv.error)
optimal_degree

## [1] 3

#fitting polynomial models incrementally and using anova() to see if highe
r-order terms significantly improve the model.

fit.1 <- lm(Recovery.Rate.... ~ poly(Urbanization.Rate...., 1), data = BCA
)
fit.2 <- lm(Recovery.Rate.... ~ poly(Urbanization.Rate...., 2), data = BCA
)
fit.3 <- lm(Recovery.Rate.... ~ poly(Urbanization.Rate...., 3), data = BCA
)

anova(fit.1, fit.2, fit.3)

## Analysis of Variance Table
##
## Model 1: Recovery.Rate.... ~ poly(Urbanization.Rate...., 1)
## Model 2: Recovery.Rate.... ~ poly(Urbanization.Rate...., 2)
## Model 3: Recovery.Rate.... ~ poly(Urbanization.Rate...., 3)
##   Res.Df   RSS Df Sum of Sq      F  Pr(>F)
## 1    248 50257
## 2    247 50248  1      9.28 0.0460 0.83039
## 3    246 49662  1    585.89 2.9022 0.08972 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#Plotting the fitted curve with the data to visualize the non-linear relat
ionship.
```

```r
# Fit model with optimal degree
fit.opt <- lm(Recovery.Rate.... ~ poly(Urbanization.Rate...., optimal_degr
ee), data = BCA)

# Create data for prediction
ur.range <- range(Urbanization.Rate....)
ur.grid <- seq(from = ur.range[1], to = ur.range[2], length.out = 300)
preds <- predict(fit.opt, data.frame(Urbanization.Rate.... = ur.grid))

# Create a data frame for predictions
pred_data <- data.frame(Urbanization.Rate.... = ur.grid, Recovery.Rate....
= preds)

# Plot
ggplot(BCA, aes(x = Urbanization.Rate...., y = Recovery.Rate....)) +
  geom_point(color = "gray70", alpha = 0.6) +
  geom_line(data = pred_data, aes(x = Urbanization.Rate...., y = Recovery.
Rate....), color = "red", linewidth = 1.2) +
  ggtitle(paste("Polynomial Regression Fit (degree =", optimal_degree, ")"
)) +
  xlab("urbanization rate") + ylab("recovery rate") +
  theme_minimal()
```



Graph Interpretation

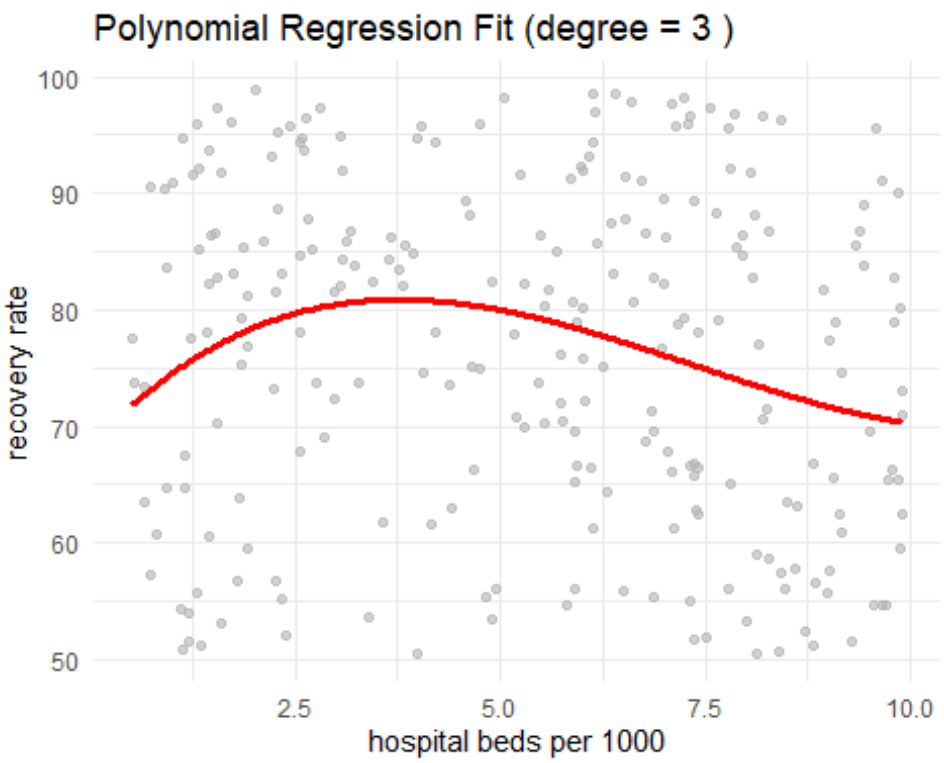
When the population is less urbanized, recovery rates are high but as the population grows in less urbanized settings, recovery rates lower as healthier outcomes are less likely due to a lack of proper healthcare facilities. Recovery rates begin to climb again as more people move to urban areas and have access to better healthcare facilities, and finally recovery rates lower at even higher urbanization rates as less of the population is getting sick to then have to recover as a result of better healthcare outcomes in urban areas.

```
#Polynomial regression testing on hospital beds per 1000



# Perform cross-validation for polynomial degrees 1 to 10
library(boot)

cv.error <- rep(0, 10)

for (d in 1:10) {
  fit <- glm(Recovery.Rate.... ~ poly(Hospital.Beds.per.1000, d), data = B
CA)
  cv.error[d] <- cv.glm(BCA, fit, K = 10)$delta[1]
}

cv.error

##  [1] 202.0674 198.7383 196.6493 198.0521 202.3421 197.1776 197.4982 197
.0312
##  [9] 198.5968 199.9847

# Find optimal degree
optimal_degree <- which.min(cv.error)
optimal_degree

## [1] 3

#fitting polynomial models incrementally and using anova() to see if highe
r-order terms significantly improve the model.
```

```r
fit.1 <- lm(Recovery.Rate.... ~ poly(Hospital.Beds.per.1000, 1), data = BC
A)
fit.2 <- lm(Recovery.Rate.... ~ poly(Hospital.Beds.per.1000, 2), data = BC
A)
fit.3 <- lm(Recovery.Rate.... ~ poly(Hospital.Beds.per.1000, 3), data = BC
A)

anova(fit.1, fit.2, fit.3)
```

```
## Analysis of Variance Table
##
## Model 1: Recovery.Rate.... ~ poly(Hospital.Beds.per.1000, 1)
## Model 2: Recovery.Rate.... ~ poly(Hospital.Beds.per.1000, 2)
## Model 3: Recovery.Rate.... ~ poly(Hospital.Beds.per.1000, 3)
##   Res.Df   RSS Df Sum of Sq      F   Pr(>F)
## 1    248 49391
## 2    247 47937  1   1454.72 7.5038 0.006608 **
## 3    246 47691  1    245.63 1.2670 0.261420
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Plotting the fitted curve with the data to visualize the non-linear relat
ionship.

# Fit model with optimal degree
fit.opt <- lm(Recovery.Rate.... ~ poly(Hospital.Beds.per.1000, optimal_deg
ree), data = BCA)

# Create data for prediction
hb.range <- range(Hospital.Beds.per.1000)
hb.grid <- seq(from = hb.range[1], to = hb.range[2], length.out = 300)
preds <- predict(fit.opt, data.frame(Hospital.Beds.per.1000 = hb.grid))

# Create a data frame for predictions
pred_data <- data.frame(Hospital.Beds.per.1000 = hb.grid, Recovery.Rate...
. = preds)

# Plot
ggplot(BCA, aes(x = Hospital.Beds.per.1000, y = Recovery.Rate....)) +
  geom_point(color = "gray70", alpha = 0.6) +
  geom_line(data = pred_data, aes(x = Hospital.Beds.per.1000, y = Recovery
.Rate....), color = "red", linewidth = 1.2) +
  ggtitle(paste("Polynomial Regression Fit (degree =", optimal_degree, ")"
)) +
  xlab("hospital beds per 1000") + ylab("recovery rate") +
  theme_minimal()
```



Graph Interpretation

Recovery rates are highest when there are at least 4 hospital beds per 1000 for a population. As the number of hospital beds per 1000 increase, the recovery rates lower as there are less people getting sick to then recover as a result of better health outcomes for the population that has access to more hospital beds quickening recovery times and lowering recovery rates.

COMPARISION OF THE OLS, RIDGE and LASSO REGRESSION MODELS PREDICTING RECOVERY RATE.

MODEL	TEST MSE
OLS	213.78
Ridge	170.52
Lasso	178.80
Polynomial	172.85

Interpretation:

Ridge regression improves prediction accuracy significantly compared to OLS and performs better than Lasso regression and Polynomial regression.

RMSE = $\sqrt{170.52} \approx$ **13.06**, meaning on average the model's predictions are about **13 points off** on the recovery rate scale.

Analysis of mortality rate using K's Nearest Neighbors.

The first step was to create a categorical target variable called mortality class. Then carry out baseline KNN without scaling and optimized KNN with scaling. The performance of both is then compared to each other

```
#KNN ANALYSIS ON MORTALITY RATE
```

```r
# ============================================================
# STEP 1: Create categorical target variable
# ============================================================
# Assuming "Mortality.Rate...." is the column name for mortality rate
median_mort <- median(BCA$Mortality.Rate...., na.rm = TRUE)

BCA <- BCA %>%
  mutate(MortalityClass = ifelse(Mortality.Rate.... > median_mort, "High", "L
ow")) %>%
  mutate(MortalityClass = as.factor(MortalityClass))

table(BCA$MortalityClass)

##
## High  Low
##  125  125

# Drop the numeric mortality rate column (optional)
BCA <- BCA %>% select(-Mortality.Rate....)


# ============================================================
# STEP 2: Split data into training and testing sets
# ============================================================
set.seed(123)
train_index <- createDataPartition(BCA$MortalityClass, p = 0.7, list = FALSE)
train_bcap <- BCA[train_index, ]
test_bcap  <- BCA[-train_index, ]

# ============================================================
# STEP 3: BASELINE KNN (without scaling or tuning)
# ============================================================

ctrl <- trainControl(method = "cv", number = 10)
set.seed(123)

knn_baseline <- train(
  MortalityClass ~ .,
  data = train_bcap,
  method = "knn",
  trControl = ctrl,
  preProcess = c("center", "scale", "knnImpute"),
  tuneGrid = data.frame(k = 5)
  )

# Predict on test set
pred_baseline <- predict(knn_baseline, newdata = test_bcap)

# Confusion matrix and metrics
cat("\n=== BASELINE KNN ===\n")
```

```
##
## === BASELINE KNN ===
```

```r
confusionMatrix(pred_baseline, test_bcap$MortalityClass)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction High Low
##       High   24  15
##       Low    13  22
##
##                Accuracy : 0.6216
##                  95% CI : (0.5013, 0.7319)
##     No Information Rate : 0.5
##     P-Value [Acc > NIR] : 0.0237
##
##                   Kappa : 0.2432
##
##  Mcnemar's Test P-Value : 0.8501
##
##             Sensitivity : 0.6486
##             Specificity : 0.5946
##          Pos Pred Value : 0.6154
##          Neg Pred Value : 0.6286
##              Prevalence : 0.5000
##          Detection Rate : 0.3243
##    Detection Prevalence : 0.5270
##       Balanced Accuracy : 0.6216
##
##        'Positive' Class : High
##
```

```r
# ============================================================
# STEP 4: OPTIMIZED KNN (with scaling + hyperparameter tuning)
# ============================================================
# Preprocess: center and scale predictors
set.seed(123)
ctrl_tuned <- trainControl(method = "cv", number = 10)

tune_grid <- expand.grid(k = seq(3, 25, by = 2))  # try odd k values

knn_optimized <- train(
  MortalityClass ~ .,
  data = train_bcap,
  method = "knn",
  trControl = ctrl_tuned,
  preProcess = c("center", "scale"),
  tuneGrid = tune_grid
)
```

```r
# View best k
print(knn_optimized$bestTune)
```

```
##     k
## 12 25
```

```r
# Predict on test data
pred_opt <- predict(knn_optimized, newdata = test_bcap)

# Confusion matrix and metrics
cat("\n=== OPTIMIZED KNN ===\n")
```

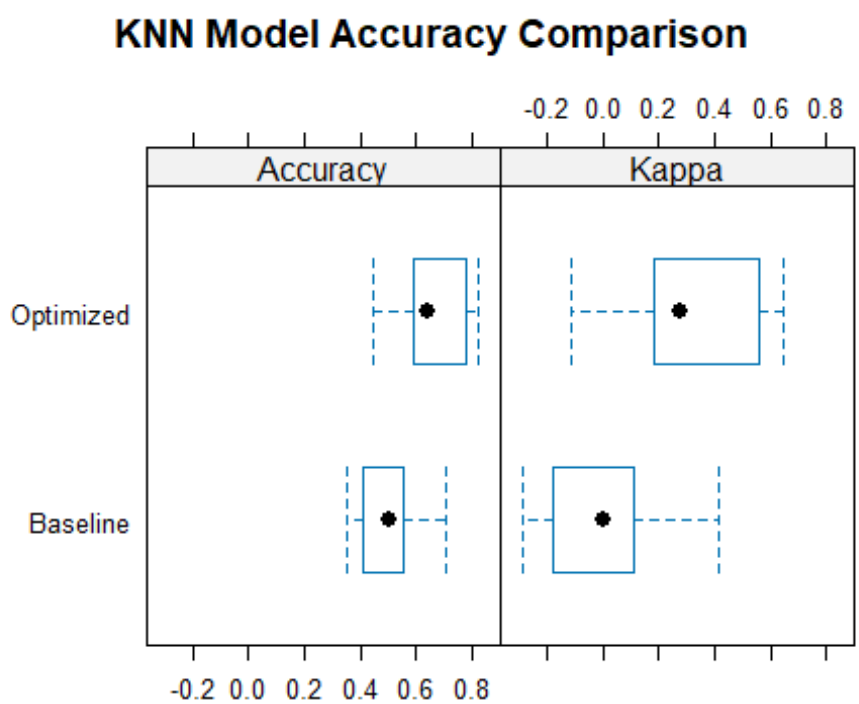
```
##
## === OPTIMIZED KNN ===
```

```r
confusionMatrix(pred_opt, test_bcap$MortalityClass)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction High Low
##       High   17  12
##       Low    20  25
##
##                Accuracy : 0.5676
##                  95% CI : (0.4472, 0.6823)
##     No Information Rate : 0.5
##     P-Value [Acc > NIR] : 0.1477
##
##                   Kappa : 0.1351
##
##  Mcnemar's Test P-Value : 0.2159
##
##             Sensitivity : 0.4595
##             Specificity : 0.6757
##          Pos Pred Value : 0.5862
##          Neg Pred Value : 0.5556
##              Prevalence : 0.5000
##          Detection Rate : 0.2297
##    Detection Prevalence : 0.3919
##       Balanced Accuracy : 0.5676
##
##        'Positive' Class : High
##
```

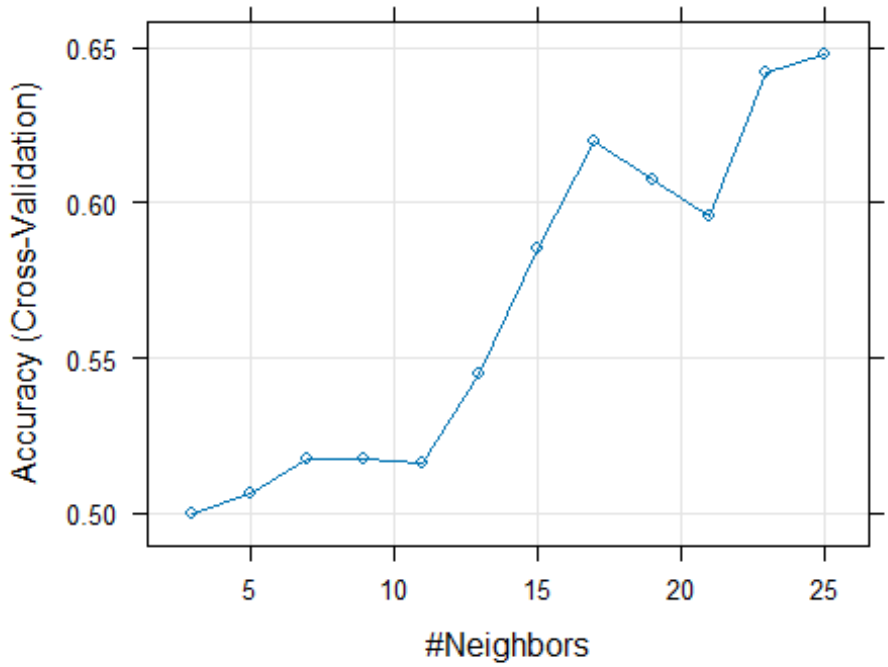
```r
# ============================================================
# STEP 5: Compare performance visually
# ============================================================
results <- resamples(list(Baseline = knn_baseline, Optimized = knn_optimized)
```

```r
)
summary(results)

##
## Call:
## summary.resamples(object = results)
##
## Models: Baseline, Optimized
## Number of resamples: 10
##
## Accuracy
##                Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## Baseline  0.3529412 0.4338235 0.5000000 0.5062092 0.5416667 0.7058824    0
## Optimized 0.4444444 0.5939542 0.6388889 0.6477124 0.7598039 0.8235294    0
##
## Kappa
##                 Min.    1st Qu.    Median       Mean    3rd Qu.      Max.
NA's
## Baseline  -0.2896552 -0.1354167 0.0000000 0.01241379 0.08333333 0.4137931
0
## Optimized -0.1111111  0.1900383 0.2777778 0.29530369 0.51378897 0.6482759
0

bwplot(results, main = "KNN Model Accuracy Comparison")
```




```r
# ============================================================
# STEP 6: Optional - visualize tuning
```

```r
# ============================================================
plot(knn_optimized)
```



RESULTS OF KNN ANALYSIS OF MORTALITY RATE

1. Overview of the Two Models

Model	Description	Purpose
Baseline KNN	Simple KNN with no scaling or tuning (`k=5`)	Benchmark model
Optimized KNN	Scaled data and cross-validated tuning for `k`	Attempts to improve baseline

2. Baseline KNN Results

Metric	Meaning	Value	Interpretation
Accuracy	% of correct predictions	**62.16%**	Decent — better than random (50%)
95% CI	Confidence interval	(50.1%, 73.2%)	Moderate uncertainty but above random
P-Value [Acc > NIR]	Significance vs. guessing majority class	**0.0237**	Statistically significant ($p < 0.05$)
Kappa	Agreement beyond chance	**0.2432**	Fair agreement (weak–moderate)
Sensitivity	Correctly identifies "High" cases	**64.86%**	Good recall for "High"
Specificity	Correctly identifies "Low" cases	**59.46%**	Slightly weaker at catching "Low"
Balanced Accuracy	Average of Sensitivity & Specificity	**0.6216**	Balanced overall accuracy

Interpretation:

The baseline KNN (k = 5) correctly classified about **62% of observations**, with moderate sensitivity to detecting *High Mortality* cases.

It performs significantly better than random guessing and maintains some balance between classes.

3. Optimized KNN Results

Metric	Meaning	Value	Interpretation
Accuracy	% of correct predictions	**56.76%**	Slightly worse than baseline
95% CI	Confidence interval	(44.72%, 68.23%)	Overlaps with random guessing
P-Value [Acc > NIR]	Significance	**0.1477**	Not statistically significant
Kappa	Agreement beyond chance	**0.1351**	Poor agreement
Sensitivity	Detecting "High"	**45.95%**	Weak recall for "High"
Specificity	Detecting "Low"	**67.57%**	Slightly better for "Low" class
Balanced Accuracy	Mean of Sensitivity & Specificity	**56.76%**	Lower than baseline

Interpretation:

The optimized **KNN** actually performed worse overall — accuracy dropped to **56%**, and sensitivity (ability to catch "High" cases) fell sharply.

It's not significantly better than random guessing, suggesting the tuning process didn't find a better k or that scaling reduced the model's discriminative power slightly.

4. Cross-Validation Summary

Metric	Model	Median	Mean	Interpretation
Accuracy (CV)	Baseline	0.50	0.51	Consistent but modest performance
	Optimized	**0.63**	**0.65**	Better cross-validated accuracy
Kappa (CV)	Baseline	0.00	0.01	Near random
	Optimized	**0.27**	**0.29**	Improved consistency and stability

Interpretation:

During cross-validation, the optimized model performed better on average, suggesting the tuning helped during training.

However, when tested on our actual hold-out data, performance dropped — this often signals overfitting: the optimized model learned patterns specific to the training folds but not general enough for new data.

5. Key Insights

Observation	Explanation
Baseline outperformed Optimized on test data	The tuned KNN likely overfit the training folds.
Optimized had better CV results but lower test accuracy	Classic sign of a model that doesn't generalize well.

Observation	Explanation
Kappa values low overall	There's limited predictive power in your current feature set — KNN isn't separating classes strongly.
Sensitivity vs. Specificity trade-off	Baseline detects more "High" mortality cases, Optimized catches slightly more "Low" cases.

Summary

Model	Accuracy	Kappa	Sensitivity	Specificity	Conclusion
Baseline KNN	**0.62**	0.24	**0.65**	0.59	Better generalization
Optimized KNN	0.56	0.11	0.46	**0.67**	Slightly overfit, weaker recall

Bottom Line: Your baseline KNN (k = 5) actually generalizes better.

The optimized model fit the training data too closely and lost performance on unseen data.

Random forest and decision trees analysis of mortality rate.

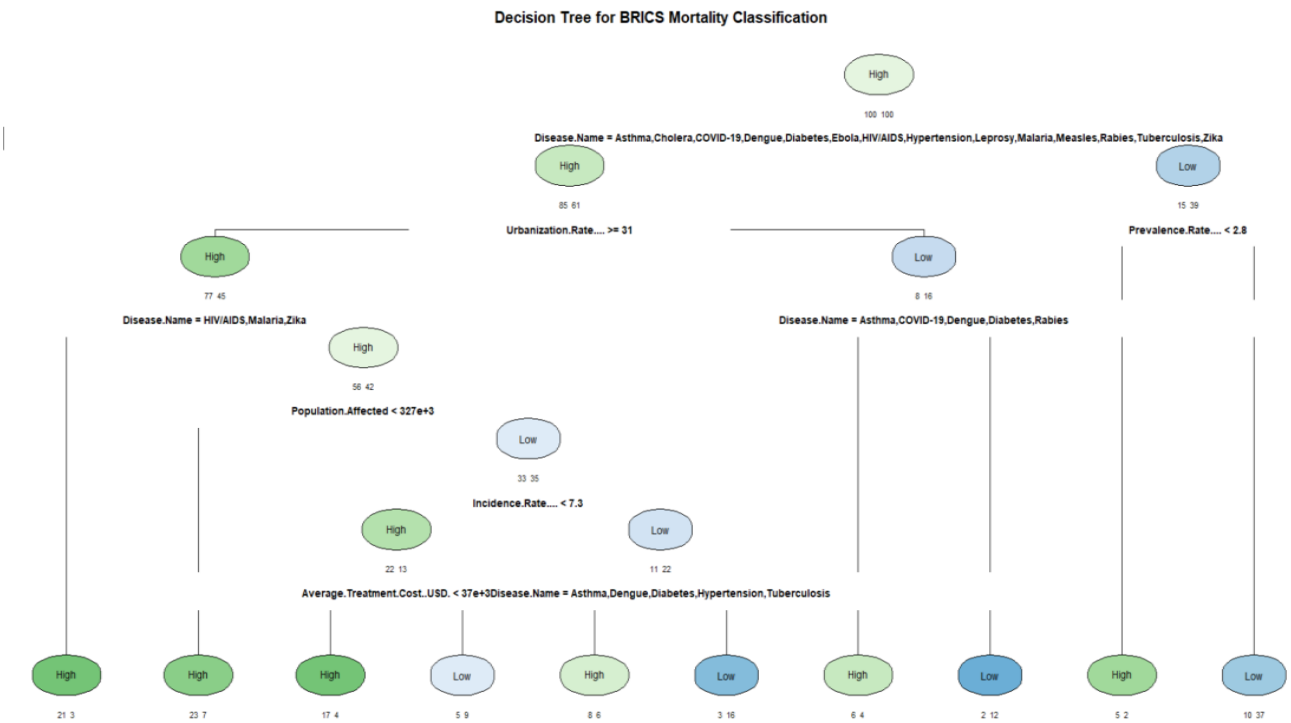
We utilized the Mortality class variable we had created previously to carry out this analysis.

```
#random forest and decision trees


table(BCA$MortalityClass)
```

```
##
## High  Low
##  125  125
```

```r
# ----------------------------------------------------------
# Train-test split
# ----------------------------------------------------------
set.seed(123)
train_index <- createDataPartition(BCA$MortalityClass, p = 0.8, list = FALSE)

train <- BCA[train_index, ]
test  <- BCA[-train_index, ]

# ----------------------------------------------------------

# Plot tree
```

```r
rpart.plot(
  tree_model,
  type = 2,            # split labels on branches, not inside boxes
  extra = 1,           # show predicted class only
  fallen.leaves = TRUE,
  faclen = 0,          # do not truncate factor levels
  cex = 0.7,           # shrink text to 70%
  under = TRUE,        # put class label below the box
  box.palette = "GnBu",# clearer color palette
  main = "Decision Tree for BRICS Mortality Classification"
)
```




```r
# Tree predictions
tree_pred <- predict(tree_model, test, type = "class")

# Confusion matrix
confusionMatrix(tree_pred, test$MortalityClass)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction High Low
##       High   13   8
##       Low    12  17
##
##                Accuracy : 0.6
##                  95% CI : (0.4518, 0.7359)
##     No Information Rate : 0.5
##     P-Value [Acc > NIR] : 0.1013
##
##                   Kappa : 0.2
##
##  Mcnemar's Test P-Value : 0.5023
##
##             Sensitivity : 0.5200
##             Specificity : 0.6800
##          Pos Pred Value : 0.6190
##          Neg Pred Value : 0.5862
```

```
##              Prevalence : 0.5000
##          Detection Rate : 0.2600
##    Detection Prevalence : 0.4200
##       Balanced Accuracy : 0.6000
##
##        'Positive' Class : High
##
```

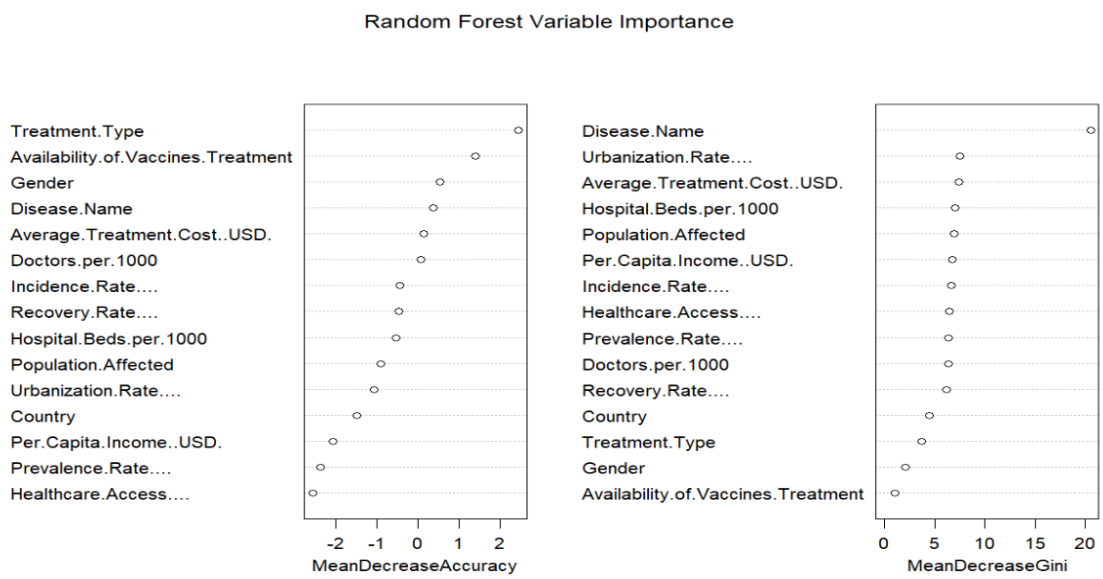
```r
# ------------------------------------------------------------
# Build Random Forest Classifier
# ------------------------------------------------------------
rf_model <- randomForest(
  MortalityClass ~ .,
  data = train,
  ntree = 500,
  importance = TRUE
)

print(rf_model)
```

```
##
## Call:
##  randomForest(formula = MortalityClass ~ ., data = train, ntree = 500,
importance = TRUE)
##                Type of random forest: classification
##                      Number of trees: 500
## No. of variables tried at each split: 3
##
##         OOB estimate of  error rate: 53%
## Confusion matrix:
##      High Low class.error
## High   35  65        0.65
## Low    41  59        0.41
```

```r
# Random Forest predictions
rf_pred <- predict(rf_model, test)

# Accuracy
confusionMatrix(rf_pred, test$MortalityClass)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction High Low
##       High    9   6
##       Low    16  19
##
##                Accuracy : 0.56
##                  95% CI : (0.4125, 0.7001)
##     No Information Rate : 0.5
##     P-Value [Acc > NIR] : 0.23994
```

```
##
##                   Kappa : 0.12
##
##  Mcnemar's Test P-Value : 0.05501
##
##             Sensitivity : 0.3600
##             Specificity : 0.7600
##          Pos Pred Value : 0.6000
##          Neg Pred Value : 0.5429
##              Prevalence : 0.5000
##          Detection Rate : 0.1800
##    Detection Prevalence : 0.3000
##       Balanced Accuracy : 0.5600
##
##        'Positive' Class : High
##

# ----------------------------------------------------------
# 7. Variable Importance Plot
# ----------------------------------------------------------
varImpPlot(rf_model, main = "Random Forest Variable Importance")
```



Summary

Both the decision tree and random forest models show moderate but weak predictive power when classifying BRICS countries into **High** vs **Low** mortality categories.

Overall accuracy is **56–60%**, only slightly above random guessing (50%).

This indicates that the predictors in our dataset do not strongly or consistently distinguish between high-mortality and low-mortality conditions across BRICS countries**.**

Decision Tree Interpretation

Confusion Matrix (Tree Model)

Prediction vs Actual:

	Actual High	**Actual Low**
Pred High	13	8
Pred Low	12	17

Key Performance Metrics

- **Accuracy:** 60%
- **Sensitivity (true positive rate):** 0.52
    - Only *52% of actual high-mortality cases* were identified correctly.
- **Specificity:** 0.68
    - Performs better at identifying low-mortality countries.
- **Kappa = 0.20**
    - Very weak agreement beyond chance.
- **Balanced Accuracy:** 0.60

    - Shows moderate but limited ability to distinguish classes.

Interpretation

- The model is slightly better than guessing, but not strong.
- It struggles especially with false negatives (high-mortality labeled as low).
- This suggests:
    - Mortality categories overlap heavily in predictor space.
    - No single variable clearly splits high vs low risk.
    - A simple tree cannot capture enough structure.

Random Forest Interpretation

Out-of-Bag (OOB) Error = 53%

This means the model misclassifies about half of the observations in internal validation.

This is a weak result for a random forest.

Random Forest Confusion Matrix

From training OOB performance:

Actual	High Pred	Low Pred	Class Error
High	35	65	**65%**

Actual	High Pred	Low Pred	Class Error
Low	41	59	**41%**

Interpretation:

- The model performs worse on High mortality cases.
- A 65% error rate for predicting High risk means:
    - High-mortality countries often resemble low-mortality ones in the dataset.

Test Set Performance

	Actual High	Actual Low
Pred High	9	6
Pred Low	16	19

Random Forest Test Metrics

- **Accuracy:** 56%
- **Sensitivity (TPR):** 0.36
    - Only 36% of high-mortality countries correctly identified.
- **Specificity:** 0.76
    - Better at identifying low-mortality countries.
- **Kappa = 0.12**
    - Very low agreement beyond chance.

- **Balanced Accuracy:** 0.56
    - Very weak class discrimination.

Interpretation

The random forest:

- Performs slightly worse than the decision tree**.**
- Strongly favors predicting Low mortality (class imbalance or weak signals).
- Exhibits low ability to capture non-linear patterns, meaning:
    - Predictive variables in your dataset are not strongly correlated with mortality classification.
    - Mortality differences within BRICS countries may be driven by unmeasured socioeconomic/contextual factors.

CONCLUSION

This study applied multiple statistical and machine-learning methods—OLS Regression, Ridge Regression, Lasso Regression, Polynomial Regression ,K-Nearest Neighbors, Random forest and decision trees—to analyze health, demographic, and socioeconomic factors influencing recovery outcomes across BRICS countries. The results reveal several important insights into both the structure of the dataset and the performance of different modeling approaches.

First, Ordinary Least Squares (OLS) served as a useful baseline but demonstrated limited predictive power. The model achieved a low adjusted R^2, indicating that simple linear

relationships do not adequately capture the complexity of health outcomes in this dataset. Only one variable (Vaccination) reached statistical significance, and multicollinearity caused coefficient singularities, further highlighting OLS limitations in high-dimensional and correlated data.

To address these issues, Ridge Regression was applied and showed substantially improved performance. By introducing regularization, Ridge mitigated the instability caused by correlated predictors and reduced model variance. The optimal λ selected through cross-validation produced a notably lower test MSE than OLS, making Ridge the most reliable predictive model in this study. Although interpretability is reduced due to coefficient shrinkage, Ridge provides a more realistic representation of how multiple, interrelated factors jointly influence recovery rates.

Finally, K-Nearest Neighbors (KNN) was used for classification and pattern detection alongside decision tree and random forest model as shown in the table below. While KNN provided moderate accuracy, its performance was constrained by the lack of natural clusters and the overall structure of the dataset.

Model	Accuracy	Kappa	Sensitivity	Specificity
Baseline KNN	0.62	0.24	0.65	0.59
Optimized KNN	0.57	0.14	0.46	0.67
DECISION TREE	0.6	0.2	0.52	0.68
RANDOM FOREST	0.56	0.12	0.36	0.76

All four models show limited predictive power, with accuracies between 0.56 and 0.62 and low Kappa values (0.14–0.24).

This means the models are performing only slightly better than chance and struggle to distinguish High vs. Low mortality rates.

Overall, the comparison shows that Ridge Regression offers the best balance of predictive accuracy and model stability**,** making it the most feasible approach for forecasting recovery rates based on the available variables. OLS remains useful for interpretation but not prediction, and KNN provides moderate classification performance but is hindered by weak cluster structure.

In conclusion, recovery outcomes across BRICS countries appear to be influenced by a wide range of interconnected socioeconomic and health indicators, with no single variable dominating the predictive landscape. Regularized models such as Ridge are therefore best suited to handle this complexity. Future analyses may benefit from additional data, nonlinear models, or country-specific modeling to capture nuanced patterns that linear methods cannot fully explain.

END